Instructor:
Email:
Office hours:

I. Course Description

The study of equations, functions and graphs, reviewing linear and quadratic functions, and concentrating on polynomial, rational, exponential and logarithmic functions. Emphasizes algebraic problem solving skills and graphical representation of functions.

II. Student Learning Outcomes

Students will build on their knowledge of polynomial, rational, absolute value, radical, exponential and logarithm functions in the following contexts:

1. **Functions and Function Notation:** Use function notation; perform function arithmetic, including composition; find inverse functions; determine the domain and range of functions algebraically; find the difference quotient of a function; find the average rate of change of a function over an interval.
2. **Representation of Functions:** Identify functions and their transformations given in algebraic, graphical, numerical, and verbal representations, and explain the connections among these representations.
3. **Graphs of Functions:** Graph and interpret key feature of functions, e.g., intercepts, leading term, end behavior, asymptotes, domain and range.
4. **Relationship Between Equations and Graphs of Functions:** Solve equations algebraically to answer questions about graphs, and use graphs to estimate solutions to equations.
5. **Applications:** Solve contextual problems by identifying the appropriate type of function given the context and creating a formula based on the information given.
6. **Communication:** Communicate mathematical information using proper notation and verbal explanations.

III. Prerequisites

Math 1215XYZ or ACT Math ≥ 22 or SAT Math Section ≥ 540 or ACCUPLACER Next-Generation Advanced Algebra and Functions: 239-248.

Note: College Algebra builds on concepts learned in Intermediate Algebra. If it has been a while since you last took a mathematics class, you will probably need to review in order to be successful in College Algebra. Also, Mathematics or Statistics coursework dating back more than five years cannot automatically be counted as fulfillment of a prerequisite. Students with coursework dating back more than five years should take the placement exam offered through the University of New Mexico Testing Center to determine which Mathematics or Statistics courses to register for based on their skill level.

IV. Texts and Materials

- **Textbook and Online HW:** The textbook is COLLEGE ALGEBRA with CoRequisite Support, 1st edition, Julie Miller and Donna Gerken. The textbook and the online HW will be accessed through ALEKS with a link in Canvas.
- **Calculator:** For some application problems, a scientific calculator may be useful, especially when you are working with exponential and logarithmic functions. **However, NO calculator will be allowed on any exams.**
- **Other Materials:**
  - **A Notebook.** You will want a notebook to take notes while you read the textbook. You will also want a notebook for your work as you are doing your HW problems. Keeping well organized notes is extremely important to your success in this class. A hole puncher and a stapler are also useful tools in organizing your notes, worksheets, quizzes, and any handouts.
  - **Computer, Internet Access, Webcam:** All students should have access to a computer with internet access to do their online HW

- **Scanning App:** You may, from time to time, need to scan your work. A scanning app works much better than just a photo from your cell phone. There are many free scanning apps available for smart phones. These apps allow you to use the camera on your phone to take black and white scans of your work and concatenate multiple pages into a single pdf file, so that you can upload your work to Canvas or email your work to your instructor. These scanning apps create files that are much easier to read than what you typically get when you simply photograph a written page. Some free apps that you can download are CamScanner, Genius Scan, Scanner Pro, TurboScan, Scannable. A search in the App Store or Google Play will show a variety of free scanning apps.

V. Course Structure

- **Learning the Material:** You will have various activities to do in this class to help you learn the material: going to class; doing homework, online or paper–and–pencil or both; actively reading the book by working through the examples. All these activities contribute to your learning and complement each other. For example, you will see examples in the textbook that are different than examples that your instructor presents. The online textbook includes embedded lectures and animations to aid you in your reading. In addition, you should expect to do problems in your homework assignments that are not covered in your lectures. The online homework system has built in tutorials to help you learn how to do those problems. Your learning will be assessed with quizzes and tests throughout the semester. You should expect to spend **at least** 9 hours per week **outside of class tim**e on this class.

- **Written Homework:** We will have weekly homework due at the last day of class each week and must be submitted online in PDF file. (I will not accept paper so please write and use scan app to submit them, if you write on ipad just submit them in PDF file as well)

- **Written Quizzes:** we will take a quiz at the beginning of class after turning in homework.

- **In class Activity:** there will be in class assignments where we get to work together and usually you will need to finish them at home.

- **Late Policy:**
  1) Homework needs to be done on due date and must submit online in PDF file and I will not accept late homework since we only have a few.
  2) Quizzes can be retaking one time during office hour.
  3) Exams may not be completed late except for documented emergencies (see statement under **Exams**.)
  4) If you are unsure about the late policy or think you may be eligible for an extension based on an extenuating circumstance, contact the instructor immediately. Granting extensions is up to the discretion of the instructor and, if granted, may incur a penalty.

Exams: There will be two written exams in this class at the end of the month. You will need to address all components of the problems, show all your work, and write out solutions using correct mathematical notation.

1) Exam dates are written in the schedule. The instructor will adhere to these dates unless something unforeseen happens, in which case the instructor will announce the new dates as soon as possible.
2) Students **may not** use a calculator or any notes, note cards, books, or other resources during the exam.
3) Students **may not** discuss the exam with anyone other than the instructor.
4) Exams cannot be rescheduled, except in documented emergencies. If you know you are going to miss an exam because of a documented emergency (e.g. surgery), or an NCAA sporting event, or military deployment, you must make prior arrangements with your instructor to reschedule the exam. Please let your instructor know as soon as you find out.
5) If you miss an exam due to an emergency or illness, you must provide documentation of the emergency (doctor's note, police report, etc.), and email the instructor within 24 hours of the missed exam. Make-up exams after the regularly scheduled exam date should be taken as soon as possible and within two business days.
6) Do not schedule a personal trip during exams as you will not be given a makeup.

VI. Grading

Your grade will be determined based on your performance on the following:

Source	Percentage
Written HW	10%
Quizzes	10%
In class activity	10%
ALEKS	10%
2 Unit Exams	60%
Total	100%

How Grades Are Determined:

98–1	A+	87–89	B+	77–79	C+	67–69	D+
93–97	A	83–86	B	70–76	C	60–66	D
90–92	A-	80–82	B-			Below 60	F

There will be extra credit but it will depend on opportunities given in class.
To get full credit on graded work students must address all mathematical components presented by the problem, showing all steps and calculations. The use of proper notation, well-structured procedures, and legibility will be taken into account when assigning points.

VII. Get Help!

We, as math faculty, do our best to present the material in a clear way and our lectures or explanations might make it seem that the material is easy. But learning math is not always easy, it can take a lot of work and effort. It is totally normal to sometimes get stuck on a problem. Learning to embrace the process of struggling and working through a solution is just as important as figuring out the solution itself.

I wants you to succeed and my office hours are offered to help you through the learning process. During office hours I can work with you on questions, clarify concepts, and work more examples. Take advantage of my office hours. I am here to help you!

In addition to getting help from your instructor, there is extra help available at:

- **CAPS:** Center for Academic Program Support. CAPS is the learning assistance program available to UNM students enrolled in undergraduate classes. CAPS is located on the 3rd floor of Zimmerman Library. Phone Number: (505) 277-7205, Website: https://caps.unm.edu.
- **Student Health and Counseling Resources: :** SHAC provides quality health and counseling services (including resources for test anxiety) to all UNM students to foster student success. Phone Number: (505) 277–3136. Website: http://shac.unm.edu/services/mental-health/index.html.
- **Accessibility Resource Center:** The Accessibility Resource Center (ARC) recognizes individuals with disabilities as an integral part of a diverse community and is committed to the provision of comprehensive resources to the University community (faculty, staff, and student) in order to create equitable, inclusive, and practical learning environments. Website: https://arc.unm.edu/index.html

VIII. Additional Departmental and University Information and Policies

Attendance and participation. Attendance (mandatory) and engagement in the class (regular homework completion, questions/comments inside and outside class, and in office hours) are necessary to succeed in this course. If you need to miss class, please let me know. Any unexplained and continued absences and lack of homework may lead to being withdrawn from the course. Please make sure to stay in touch with me in case of special circumstances that temporarily prevent you from participating as needed.

Homework. You are encouraged and welcome to work together on the homework. However, the writeup you hand in must be your own work, in your own words.

Referral to other sources outside of the material given in class (such as searching the web for answers) is strongly discouraged. It does not lead to understanding. To understand the material you have to work through it. You learn mathematics, just as you do the violin, or soccer, by practice, practice, practice. And just like playing the scales, or doing the dribbling skills, it's not necessarily always fun, but necessary. But do not bang your head in frustration!.

Spend at most 20 minutes on a problem, if you are stuck contact me or other resources above for help.

Academic dishonesty.
Academic dishonesty will be reported to the Dean of Students.

Academic dishonesty includes copying answers from other sources to complete your homework (see above) or using external sources (other than pencil and paper) to complete your exams.

Grade Mode Change and Withdrawals: Deadlines to make changes to your registration status are published by the Office of the Registrar in the schedule of classes: http://registrar.unm.edu. To change grade mode or to withdraw after the deadlines posted therein, you need to 1) talk to your instructor to fully understand your standing in the class, and then 2) meet with your advisor and discuss the best path for you to proceed, as well as all consequences for your studies. Please ask your advisor to email your instructor, with copy to you, of the final decision. For grade mode changes you may also be required to have your instructor sign a grade mode change form:
http://www.unm.edu/~unmreg/images/Forms/EnrlAuth-GradeMode.pdf , and your instructor will accommodate the change. Please note that to receive a W you need to withdraw by 5 pm on the Friday before final exams week.

Credit Hour Statement:
This is a three credit-hour course. Class meets for either three 50-minute sessions of direct instruction for fifteen weeks. Please plan for a minimum of six hours of out-of-class work (or homework, study, assignment completion, and class preparation) each week.

Accommodations: UNM is committed to providing courses that are inclusive and accessible for all participants. As your instructor, it is my objective to facilitate an accessible classroom setting, in which students have full access and opportunity. If you are experiencing physical or academic barriers, or concerns related to mental health, physical health and/or COVID-19, please consult with me after class, via email/phone or during office/check-in hours (I am not legally permitted to inquire about the need for accommodations). We can meet your needs in collaboration with the Accessibility Resource Center (https://arc.unm.edu/) at arcsrvs@unm.edu or by phone (505) 277-3506.

> Support: Contact me at lyle1205@unm.edu or in office/check-in hours and contact Accessibility Resource Center (https://arc.unm.edu/) at arcsrvs@unm.edu (505) 277-3506.

Title IX: Our classroom and our university should always be spaces of mutual respect, kindness, and support, without fear of discrimination, harassment, or violence. Should you ever need assistance or have concerns about incidents that violate this principle, please access the resources available to you on campus. Please note that, because UNM faculty, TAs, and GAs are considered "responsible employees" by the Department of Education, any disclosure of gender discrimination (including sexual harassment, sexual misconduct, and sexual violence) made to a faculty member, TA, or GA must be reported by that faculty member, TA, or GA to the university's Title IX coordinator. For more information on the campus policy regarding sexual misconduct, please see: https://policy.unm.edu/university-policies/2000/2740.html.

> Support: LoboRESPECT Advocacy Center and the support services listed on its website, the Women's Resource Center and the LGBTQ Resource Center all offer confidential services and reporting.

Respectful and Responsible Learning: We all have shared responsibility for ensuring that learning occurs safely and equitably. UNM has important policies to preserve and protect the academic community, especially policies on student grievances (Faculty Handbook D175 and D176), academic dishonesty (FH D100), and respectful campus (FH CO9). These are in the Student Pathfinder (https://pathfinder.unm.edu) and the Faculty Handbook (https://handbook.unm.edu). Please ask for help in understanding and avoiding plagiarism or academic dishonesty, which can both have very serious consequences.

> Support: Center for Academic Program Support (CAPS). Many students have found that time management workshops can help them meet their goals (consult (CAPS) website under "services").

Connecting to Campus and Finding Support: UNM has many resources and centers to help you thrive, including opportunities to get involved, mental health resources, academic support including tutoring, resource centers for people like you, free food at Lobo Food Pantry, and jobs on campus. Your advisor, staff at the resource centers and Dean of Students, and I can help you find the right opportunities for you.